

Please read the following text carefully. The questions later in the study refer to it:

Imagine we find ourselves in the year 2030. Technological advances in material development and nanotechnology made it possible for the company NanoTec to develop a new jacket known as the Nano-Pat-Parka.

The text describes properties and functionalities of the Nano-Pat-Parka:

The bio-inspired Nano-Pat-Parka consists of an ecological down inner jacket and a reliable and versatile weather protection outer jacket. The outer jacket contains cadmium, a substance that is toxic. Therefore, the Nano-Pat-Parka is not compostable.

The Nano-Pat Parka is adaptive and autonomous. It can adapt to different external and internal conditions with the help of complex sensory systems. This can result in loud, clacking noises and unexpectedly result in bulky looking appearances. Due to the smart technology and nanoparticles contained in it, the Nano-Pat-Parka is able to learn. The adaption to external conditions cannot be switched on and off. What becomes possible with continuous use of the jacket is unknown so far.

If there is friction, e.g., when wearing a backpack, the woven molecular N100pX laminate becomes stiff, this can lead to unpleasant smells. In case of contact with stones or objects, the micro-electromechanical inserts in the forearm and back area can dynamically act as protectors.

Having a self-luminous and reflective surface, the outdoor jacket is highly visible and can use solar power to be completely energy-autonomous. The Nano-Pat parka is relatively maintenance-intensive. For example, the built-in technology may start to buzz, which needs to be fixed.

By recording and transmitting data (such as the external temperature or body temperature) and GPS data, the company NanoTec aims to improve the Nano-Pat parka continuously. The data sent in this way also allows the Nano-Pat-Parka to call an ambulance completely autonomously in case of an emergency. To this end, the Nano-Pat-Parka is just a prototype but is continuously tested for its functionality.

The question is whether the properties of the Nano-Pat-Parka are an appropriate response to the changing climatic conditions around the world to provide increased protection for people living in challenging environments. The questions are about whether the benefits outweigh the costs for society and what the ethical and societal implications of the Nano-Pat-Parka are.